

PROPOSAL ANALYSIS

Machine Learning for Mental Health Prediction

Venue targeting, alignment with the PROCAST/TRIPOD+AI categorisation, gap review, dataset acquisition and codebase status

Prepared for Dr. Aminat Abiola Ajibola • Companion to repository [rogerpanel/cv/mlmh-appraisal](#)

1. Executive summary

- The reframing in the protocol is correct and necessary.** The descriptive question is occupied (Hasan et al. 2026 screened 3,320 records on essentially the same scope). The appraisal question is open at field breadth, *provided* the appraisal uses PROCAST+AI (BMJ 2025) and TRIPOD+AI (BMJ 2024) rather than the 2019/2015 instruments, and provided Paper A positions itself explicitly against Meehan et al. (*Mol Psychiatry* 2022; 308 psychiatric prediction models; 22.1 % reported calibration), which is the closest prior work and is not mentioned in the package.
- Venue.** Paper A: *npj Digital Medicine* (first choice if the yield is ≥ 100 studies and PROCAST+AI is used), *JMIR Mental Health* (second), *Journal of Clinical Epidemiology* (methodological fallback). Paper B: *JMIR AI* (first; it has just published a participant-aware validation study asking the E1 question in another domain), *IEEE JBHI* (second). The brief's primary choice, *Journal of Biomedical Informatics*, is listed as Q2 in the 2026 JCR summary found by search; keep it as a fallback for B, not as the first target for A.
- The experimental design aligns with the categorisation** once each review outcome is tied to a PROCAST(+AI) signalling question, a TRIPOD+AI item, one experiment, one module and one test (Section 4). Two items are not covered by the code and must be handled in the text: fairness by subgroup (sex is available) and pre-registration of E1–E3 (OSF; cheap).
- Two design hazards were found that the brief does not mention.** (i) The PSYKOSE control group *is* the DEPRESJON control group (32 identical files; this is why OBF-Psychiatric has 162 participants), so the obvious E2 pair leaks unless the shared controls are assigned to one side; the code detects this from the activity series and does the assignment. (ii) HYPERAKTIV's "controls" are psychiatric patients without ADHD, so DEPRESJON→HYPERAKTIV is a label-shift test, not an external validation of the same task; it is kept as a deliberately adversarial pair.
- The instruments in the workbook need re-keying.** The TRIPOD_AI sheet uses a custom 27-line numbering that does not follow the published checklist; the PROCAST sheet reproduces 2019 PROCAST. Adherence percentages will be challenged unless both are aligned to the official items.
- Codebase.** `mlmh-appraisal/` is scaffolded exactly to the brief's specification, with 26 passing tests (the subject-leakage test first), a synthetic fixture that mimics every dataset's on-disk layout, and an end-to-end run producing LaTeX tables, PDF figures, manifests and a TRIPOD+AI self-audit. **No real data could be fetched:** every dataset host is blocked by the sandbox's egress policy. Section 6 gives exact links and a step-by-step route to bring the archives back into the session.

2. Reading of the proposal package

The package comprises the project brief (paired-paper design, dataset plan, repository specification), the review protocol (overlap check, reframed question, PICOS, four search strings, PROSPERO text, workload) and a seven-sheet extraction/appraisal workbook. Its logic is sound: Paper A measures prevalence of methodological failures; Paper B measures their cost; together they form one argument. The following observations affect execution.

- **Instruments.** The brief says “PROBAST (20 signalling questions, 4 domains)”. That is the 2019 tool. PROBAST+AI (Moons, Damen, Collins et al., BMJ 2025) is the current instrument for ML models and adds explicit questions on data leakage and on AI-specific sample-size expectations; using it is both correct and a novelty claim. The brief’s “ $EPV \geq 200$ for ML (Moons et al., 2025)” must be checked against the published wording before it is quoted **[verify]**.
- **Workbook TRIPOD_AI sheet.** Item numbers 1–27 in the sheet are custom (e.g. the sheet’s item 11 is “class imbalance”, the published item 16 is; the sheet’s 19/20 are data/code availability, published item 21 is “Open science”). Re-key to the published 27 items / 52 sub-items before piloting.
- **Workbook PROBAST sheet.** Signalling question 4.1’s EPV text should carry the PROBAST+AI wording; add the PROBAST+AI leakage and transportability questions.
- **PRISMA.** The earlier draft’s flow diagram did not reconcile (14 vs 15). The repository ships a PRISMA template whose counts are computed from macros so the figure cannot disagree with the text.
- **Dataset table corrections.** HYPERAKTIV: 51 ADHD + 52 clinical controls, activity available for 85, HRV for 80, not “ADHD patients”. OBF-Psychiatric is on Zenodo (record 13754984). DAIC-WOZ: the WOZ subset used in AVEC has 142 participants with PHQ-8 in the official split; 189 is the full DAIC session count **[verify]**.
- **Sequencing.** The brief asks for the leakage test before any model code. That order was followed: `tests/test_no_subject_leakage.py` was written first and passes against synthetic fixtures; the CV runner also asserts disjointness at run time.

3. Venue targeting

Journal metrics below were read from public indexing pages found by web search on 5 September 2026 and must be re-verified in JCR/Scopus before submission.

3.1 Paper A (systematic review and critical appraisal)

Venue	Why	Risk
npj Digital Medicine (IF ≈ 18)	Publishes PROBAST critical appraisals of ML models; state-of-the-field methods critiques are core scope	Very selective; needs the full 80–200 study yield and PROBAST+AI; APC
JMIR Mental Health (IF ≈ 7.4 , Q1 Psychiatry)	Exact readership; methodological reviews of digital mental-health AI; fast; long supplements accepted	APC; less methodological prestige
Journal of Clinical Epidemiology (Q1)	Home of the PROBAST/TRIPOD community; right if the centre of gravity is the instrument (first PROBAST+AI use at breadth, EPV finding)	Readers are methodologists, not psychiatrists
Molecular / Translational Psychiatry	Meehan 2022 precedent; 2025 generalizability review	Must show what is added over Meehan: ML/digital modalities, PROBAST+AI, leakage outcome, $EPV \geq 200$
J. Biomedical Informatics (brief's primary; 2026 listing IF 5.9, Q2)	Would accept; methods-friendly	Under-sells a review; weak mental-health readership
Int. J. Medical Informatics (IF ≈ 5.0 , Q1)	Safe fallback	Lower visibility

3.2 Paper B (empirical companion)

Venue	Why	Risk
JMIR AI	Published “Participant-Aware Model Validation for Repeated-Measures Data” (2026;e87728): stratified 10-fold overestimated performance through subject-level leakage, group/nested CV gave conservative estimates. E1 is that question on mental-health data	Young journal; indexing maturing
IEEE J. Biomedical and Health Informatics (Q1)	Wearable/actigraphy ML readership; the Simula cohorts are known there	Slow; engineering presentation expected
J. Biomedical Informatics	Comfortable with cautionary empirical results; pairs with A if A goes elsewhere	Q2
npj Digital Medicine (brief communication)	Highest-impact pairing if E1–E3 effects are large and clean	Selectivity; two APCs
Computers in Biology and Medicine (IF ≈ 8.4) / AI in Medicine	Broad ML-in-medicine venues	Reviewer pool less attuned to PROCAST/TRIPOD framing

Submission strategy. Preprint A (medRxiv) at submission; submit B only when A has a preprint DOI; cross-cite; tell both editors. Do not send both to the same journal at once without saying so.

4. Alignment of the research and experimental structure with the categorisation

Every Paper A primary outcome is tied to the item that scores it, the experiment that prices it, the module that implements it and the test that enforces it.

Paper A outcome	PROBAST	TRIPOD	Paper B experiment	Code	Enforced by
External validation performed	4.8; applicability	10e, 24	E2: train on A, test on B, no refit	<code>evaluation/external.py</code>	<code>assert_cohorts_disjoint</code> ; test refuses overlapping cohorts
Calibration reported	4.7	12, 24	E3: slope, intercept, Brier, ECE, reliability curves	<code>evaluation/metrics.py</code>	<code>test_metrics.py</code> recovers known miscalibration
AUROC vs accuracy only	4.7	12	all experiments report AUROC, AUPRC, F1, MCC with accuracy	<code>metrics.py</code>	key/range test
Leakage-prone design	4.8; +AI leakage	10d, 11	E1: record-wise vs subject-wise, paired	<code>evaluation/splitters.py</code>	<code>test_no_subject_leakage.py</code> fails the build for any non-E1 use of <code>record_wise</code> reported
EPV ≥ 200	4.1	8	reported per cohort (E2 table)	<code>experiments._epv</code>	—
Class prevalence	1.2 / 3.4	22	prevalence in every metrics row	<code>metrics.py</code>	—
Resampling inside folds	4.8	11	SMOTE optional inside imblearn Pipeline	<code>models/pipelines.py</code>	<code>test_pipeline_fit_order.py</code>
Code / data availability	—	19–21	repository; manifests with SHA and checksums	<code>reporting/manifest.py</code>	—
Near-perfect performance cases	4.8	24	E1 inflation size	<code>run_e1</code>	paired subject bootstrap
Uncertainty intervals	4.8	24	subject-level BCa bootstrap	<code>evaluation/bootstrap.py</code>	subject-resampling test

Not covered by code: fairness/subgroup performance (add a sex-stratified analysis; sex exists in all three Simula cohorts) and pre-registration of the analysis plan (OSF, before real data are run). Both are manuscript obligations, and Paper B cannot score other studies on items it does not itself satisfy.

5. Gap review

5.1 Reviews to position against

- **Hasan et al. 2026**, *Front Digit Health* (doi 10.3389/fdgth.2025.1724348): descriptive taxonomy of ML methods and datasets, 2015–2024, 3,320 records; no risk-of-bias or reporting appraisal. Occupies the descriptive question completely.
- **Meehan et al. 2022**, *Mol Psychiatry* (doi 10.1038/s41380-022-01528-4): 308 clinical prediction models in psychiatry; 22.1 % (68/308) tested calibration, of which 52.9 % presented a plot and 22.1 % only Hosmer–Lemeshow; 75.6 % used some internal validation but only 11.4 % to an adequate standard. Closest prior work. Paper A adds: ML/digital-phenotyping modalities (sensor, speech, social media), PROBAST+AI and TRIPOD+AI, a leakage-design outcome, the ML EPV expectation, and the paired empirical companion.
- **Generalizability of clinical prediction models in mental health**, *Mol Psychiatry* 2025 (doi 10.1038/s41380-025-02950-0): external validation is rare **[verify]** exact proportions.
- Narrow PROBAST reviews: adolescent EHR crisis prediction (5 studies, 2025); psychosis prediction in at-risk mental state (2/48 models at low risk of bias); ADHD individualized prediction (*Mol Psychiatry* 2024); social-media ML biases (arXiv 2410.16204). None at field breadth.
- **Responsible Evaluation of AI for Mental Health**, ACL 2026 (arXiv 2602.00065): 135 NLP papers, generic metrics, little clinician involvement. Useful for the discussion, not an appraisal.

5.2 What the benchmarks actually show on the chosen cohorts

Study (cohort)	Split unit	Best re-ported	Comment
Garcia-Ceja 2018 (DE-PRESJON dataset paper)	leave-one-patient-out; day features; RF, DNN; SMOTE compared	F1 0.73, MCC 0.44	The honest subject-wise baseline; no calibration, no external validation
2D-CNN on time series, 2022 (PMC9495338)	per-window	accuracy 76.7%	no calibration
Transfer-learning screening tool, arXiv 2303.07847	“modified leave-one-out”	accuracy 0.96	split modification is the design the review flags
Explainable XGBoost, JMIR Ment Health 2025;e72038	not stated in abstract [verify]	accuracy 84.9% binary, 85.9% multi-class	day-wise k-fold would be record-wise leakage
CoAtNet/ViT on actigraphy images, arXiv 2512.00103	three-fold subject-wise	CoAtNet-Tiny best	one of few stating subject-wise
Jakobsen 2020 (PSYKOSE dataset paper)	leave-one-patient-out	[verify]	baseline
Multi-branch DL (Soft Computing 2024); DL+XAI (BSPC 2024) (PSYKOSE)	day-level	accuracy ≈ 0.94	day-level $\Rightarrow$ record-wise unless stated [verify]
Night-time signal classification 2022 (PMC9318635; DEPRESJON+PSYKOSE)	—	—	check double-counting of the shared controls
Hicks 2021 (HYPERAKTIV dataset paper)	10-fold on features	modest	code on GitHub simula/hyperaktiv
OBF-Psychiatric, Sci Data 2025	multi-class baselines	—	canonical harmonised copy; single control group
DAIC-WOZ audits: ICMI 2025 companion (doi 10.1145/3747327.3763034); arXiv 2605.23977; arXiv 2404.14463	official split	—	“most DAIC-WOZ classifiers are invalid”; CV winner ranks 20th on test, test winner 41st by CV; therapist prompts leak labels

5.3 The real gaps, as testable claims

- Unit-of-analysis leakage is common and its size on these cohorts is unmeasured.** No paper reports the paired record-wise vs subject-wise difference on the same data; the dataset authors’ subject-wise baseline (F1 0.73) sits far below later day-level claims (>0.9). (E1)
- Cross-cohort external validation of actigraphy models has not been reported, and the obvious pair is contaminated** (shared controls). The shared-control assignment table is itself a reportable finding. (E2)
- Calibration is never reported for sensor-based mental-health models.** No reliability curve for any DEPRESJON/PSYKOSE model exists in the literature. (E3)
- Sample size is far below any EPV guidance and window counts hide it.** 23 cases and ≈ 30 features give subject-level EPV <1 on DEPRESJON; hundreds of days make it look 10–20 $\times$ larger. Reporting both levels with subject-level intervals is a corrective. (E2 table)
- Reproducibility.** Follow-up papers rarely release code; none release manifests.
- Transdiagnostic and multiclass framing is under-examined;** OBF-Psychiatric enables it with a de-duplicated control group. DEPRESJON $\rightarrow$ HYPERAKTIV is a deliberate label-shift test.

6. Datasets: what could not be fetched, exact links, and how to bring them in

The session's egress policy blocks `datasets.simula.no`, `zenodo.org`, `nature.com`, `dcapswoz.ict.usc.edu`, `wwwn.cdc.gov`, `uni-siegen.de` and `dartmouth.edu` (HTTP 403 at the proxy; only PyPI and the GitHub repository are reachable). Download on your machine, keep the archives zipped, and attach them to the chat; they arrive under `/root/.claude/uploads/` and are unzipped into `data/raw/<cohort>/`, then `python -m mlmh verify-data` confirms the layout. Full instructions, expected folder layout and a checksum table are in `data/README.md`.

Dataset	Where	Terms
DEPRESJON	https://datasets.simula.no/depresjon/ (“Download” link; <code>downloads/depresjon.zip</code>)	open for research; cite MMSys 2018
PSYKOSE	https://datasets.simula.no/psykose/	open; cite CBMS 2020
HYPERAKTIV	https://datasets.simula.no/hyperaktiv/ ; code https://github.com/simula/hyperaktiv	open; cite MMSys 2021
OBF-Psychiatric	https://zenodo.org/records/13754984 (DOI 10.5281/zenodo.13754984); paper https://doi.org/10.1038/s41597-025-04384-3	open; cite Sci Data 2025
DAIC-WOZ / E-DAIC	https://dcapswoz.ict.usc.edu/ – sign the EULA form on the page with an academic e-mail; E-DAIC (AVEC 2019) through the same route	academics only; no redistribution; weeks
NHANES DPQ (PHQ-9)	https://wwwn.cdc.gov/nchs/nhanes/ – DPQ_J.XPT (2017–18), P_DPQ.XPT (2017–Mar 2020), DPQ_L.XPT (2021–23) + DEMO	fully open
WESAD	https://ubi29.informatik.uni-siegen.de/usi/data_wesad.html (2.5 GB; also UCI ML repository)	open; cite
StudentLife	https://studentlife.cs.dartmouth.edu/dataset.html (<code>dataset.tar.bz2</code> , ≈5 GB)	open
MODMA	https://modma.lzu.edu.cn/data/index/ (application form)	on request

Order of priority: DEPRESJON and PSYKOSE first (small, enough for E1–E3), then HYPERAKTIV, then OBF-Psychiatric. Submit the DAIC-WOZ EULA now; it is the only external dependency with a lead time.

7. Codebase: what exists and what has been verified

`mlmh-appraisal/` in the `rogerpanel/cv` repository, branch `claude/mental-health-prediction-proposal-s70` implements the brief's tree: `schema.py` (window-level records carrying `subject_id`, `cohort`, `window_id`; a subject may carry only one label), one loader per cohort into that schema (delimiter and column-name sniffing; activity-series hashing to detect the same participant across cohorts), day windowing, statistical and circadian features (M10, L5, relative amplitude, IV, day/night ratio, autocorrelation, spectral summaries), a model registry (majority, logistic regression, random forest, XGBoost, MLP, optional 1D-CNN in PyTorch), imblearn pipelines, `SubjectWiseSplit/LeaveOneSubjectOut/RecordWiseSplit`, metrics with calibration slope/intercept/Brier/ECE at window and subject level, subject-level BCa bootstrap and paired bootstrap for the E1 difference, an external-validation runner that refuses overlapping cohorts and assigns shared controls, LaTeX table and PDF figure writers, run manifests, and a TRIPOD+AI self-audit generator. Tests: 26, all passing; the leakage test also scans every config on disk.

Verified end to end on a synthetic fixture that reproduces each dataset's on-disk layout (including the duplicated PSYKOSE controls and HYPERAKTIV's semicolon-separated files): the pipeline produces `paper/empirical/tables/e1_auroc_inflation.tex`, `e1_full_metrics.tex`, `e2_external_validation.tex`, `e3_calibration.tex`, the figures, manifests and `results/synthetic/tripod_a`. Every synthetic table is watermarked; none of its numbers is a result. The README's results section is regenerated by `scripts/update_readme.py` so it tracks the latest run.

8. Roadmap

1. Register Paper A (PROSPERO text supplied); submit the DAIC-WOZ EULA; download the Tier-1 archives.
2. Re-key the TRIPOD_AI sheet to the published 27 items and the PROBAST sheet to PROBAST+AI; pilot on five studies.
3. Upload the archives; run `verify-data`, `prepare`, E1–E3 with 10 seeds and 1,000 bootstrap replicates; commit `results/real/`.
4. Add the sex-stratified analysis and the OBF-Psychiatric multiclass arm; pre-register on OSF before step 3 if at all possible.
5. Run the four searches; screen in Rayyan; extract; build the PRISMA figure from the search log.
6. Freeze a tagged release; draft both manuscripts from the generated tables; preprint A; submit A, then B.

Sources consulted (web search, 5 September 2026)

- Hasan MJ et al. Front Digit Health 2026. <https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1724348/full>
- Meehan AJ et al. Mol Psychiatry 2022. <https://www.nature.com/articles/s41380-022-01528-4>
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- Hicks SA et al. HYPERAKTIV. ACM MMSys 2021. <https://dl.acm.org/doi/10.1145/3458305.3478454>
- Explainable AI for depression detection from activity data. JMIR Ment Health 2025;e72038. <https://mental.jmir.org/2025/1/e72038>
- 2D-CNN on Depresjon. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9495338/>; transfer learning screening tool <https://arxiv.org/abs/2303.07847>; ViT/CoAtNet on actigraphy images <https://arxiv.org/pdf/2512.00103>
- Participant-Aware Model Validation for Repeated-Measures Data. JMIR AI 2026. <https://ai.jmir.org/2026/1/e87728>; Beyond participant-level CV <https://arxiv.org/html/2608.26205v1>
- Impact of the choice of cross-validation techniques (subject-wise vs record-wise). <https://pmc.ncbi.nlm.nih.gov/articles/PMC8369053/>
- A multi-probe audit of clinical-interview depression detection benchmarks. <https://arxiv.org/abs/2605.23977>; Most DAIC-WOZ classifiers are invalid. <https://dl.acm.org/doi/10.1145/3747327.3763034>; therapist prompts <https://arxiv.org/pdf/2404.14463>
- Responsible Evaluation of AI for Mental Health. ACL 2026. <https://aclanthology.org/2026.acl-long.347/>
- Journal metrics: JBI <https://www.journalmetrics.org/journal/journal-of-biomedical-informatics>; IJMI <https://www.journalmetrics.org/journal/international-journal-of-medical-informatics>; JMIR family <https://www.jmir.org/announcements/182>; npj Digit Med <https://research.com/journal/npj-digital-medicine>; CBM <https://research.com/journal/computers-in-biology-and-medicine-1>
- DAIC-WOZ access <https://dcapswoz.ict.usc.edu/>; WESAD https://ubi29.informatik.uni-siegen.de/usi/data_wesad.html; StudentLife <https://studentlife.cs.dartmouth.edu/>; MODMA <https://github.com/UAIS-LANZHOU/MODMA-Dataset>; NHANES DPQ https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/DataFiles/DPQY_L_R.htm